Tulsa Tube Bending
Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel 918-446-4461
Fax 918-446-1595
Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer: DNOW

Certificate No.: 20260615

Customer PO No.: 662974020

Issue Date: 6/15/2026

TTB Job No.: 26H109149

Material Size, Grade: 8.63 inch OD x 0.322 inch w.t., API-5L-X52 PSL 2 ERW

Quantity of Bends: 6

Material Heat No(s): B416745
Bend ID Nos.: 109149-1-1 109149-1-2 109149-1-3 109149-1-4

Material Heat No(s): B416740
Bend ID Nos.: 109149-2-2 109149-2-1

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

ASME B16.49-2023

*With the following notes and exceptions:

* B16.49 SR15.1 - Bends will be in the as bent condition in accordance with ASME B16.49 SR15.1.
* SR15.8.2 Multiple heats of material used for production bends based on similar chemistries.
* Segmentable per SR15.3, 1% max ovality in the bend area.

Approved By: Nate Wallace Date: 6/15/2026

MILL TEST CERTIFICATE

AXIS PIPE AND TUBE

AXIS PIPE AND TUBE, LLC.
1451 Louis E. Mikulin Rd.
Bryan TX 77807

Customer:	Sales Order:	Invoice No.:
Industrial Piping Specialists PO Box 681270 Tulsa,OK,74158 918-437-9100	0630005397 **PO:** WP012766	1600051929 **Date of Issue:** 02/14/2025

Certificate No.:	1600051929-3
Product Description:	5L 8.625 0.322 28.58# X52BDRH5BW-20P2
Specification:	5L 46th Edition May 2019

Heat Number	Lot Number	Mat Number	Steel Making / Coil Rolling
B416745	LT25028015	333651	Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US

Mechanical Properties

Tensile Test - Strip Specimen Gauge Length 2"							
Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)
L90°	1 1/2"	65300	73800	77400	0.89	36.8	82

CVN	Temp (F) -20								Temp (F) -20										
	Body (ft-lbf)				Shear Area (%)				Weld (ft-lbf)				Shear Area (%)				Size	Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg			
	175	179	177	177	100	100	100	100	173	150	170	165	100	90	100	97	10x6.7 mm	3T90	13.3
	Lateral Expasion (in)				UB	UB	UB	UB	Lateral Expasion (in)				064	UB	UB	0.064			

Chemical Composition (Max %)

Analysis (H/P)	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results
H	0.06	0.230	0.83	0.013	0.002	0.04	0.07	0.010	0.11	0.002	0.019	0.027	0.016	0.0003	0.0020	0.0044	0.25	0.12
P	0.06	0.229	0.81	0.013	0.002	0.04	0.08	0.010	0.11	0.003	0.014	0.024	0.015	0.0003	0.0025	0.0081	0.25	0.12
P	0.05	0.228	0.81	0.014	0.002	0.04	0.08	0.010	0.11	0.003	0.014	0.023	0.014	0.0002	0.0025	0.0090	0.25	0.11

Following Controls / Tests have been satisfactorily performed:

1.- Hydrostatic Testing: 3,900.00 psi Pressure for 5.00 seconds
2: Ultrasonic Tested to N10 - passed
3: Seam Annealed min 1,600 F
4: Flattening Test Passed

Remarks:

1: This material meets:Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 02/2025
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

Acronyms:

L = Longitudinal
Y.S. = Yield Strength
E.L. = Elongation
P = Product Analysis
UB = Un-Broken
T = Transverse
U.T.S. = Ultimate Tensile Strength
H = Heat Analysis
A.C. = Acceptance Criteria
N/A = Not Applicable

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Mahir Aroza

Quality Manager

Qualification Bend Report

Tulsa Tube Bending

Customer:	DNOW	Report No.:	109149-QBR-1
Customer PO No.:	662974020	**Code of Fabrication:**	
TTB Job No.:	26H109149	ASME B16.49-2023	

Mother Pipe Information

Pipe OD (in): 8.625 Pipe Grade: API-5L-X52 PSL 2 Pipe Heat No.: B416745

Bend Information

Radius (in): 40 Angle (degrees): 69 Bending Machine ID: M2

Tangent 1 (in): 20 MIN Tangent 2 (in): 6 MIN Bend ID No.: 109149-QB-1

Essential Variables

Essential Variable	Value	Limits of Variation
Carbon Equivalent	0.120	n/a
Pipe Diameter	8.625	None Allowed
Wall Thickness (inch):	0.322	+/- 0.12 inch
Radius-to-Diameter Ratio:	5.0	+1R - 0
Coil Design:	TTB High Spec	None Allowed
Coolant Type:	Water	None Allowed
Coolant Temperature (°F):	75.7 - 77.2	+/- 25°F
Coolant Flow Rate (GPM):	7.7	+/- 10%
Weld Seam Location:	0 degrees	15° from Neutral Axis
Induction Frequency (Hz):	918	+/- 20%
Heat Treatment Holding Temp (°F):	**SR15.1 AB**	+/- 25°F
Heat Treatment Soak Time (min):	**SR15.1 AB**	+ 15 min / - 0 min

Note 1: Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

OPERATOR: DW **DATE:** 5/29/2026

Production Bend Hardness Report

Tulsa Tube Bending

Customer:	DNOW	Report No.:	109149 H-1
Customer PO No.:	662974020	Code of Fabrication:	
TTB Job No.:	26H109149	ASME B16.49 2023	

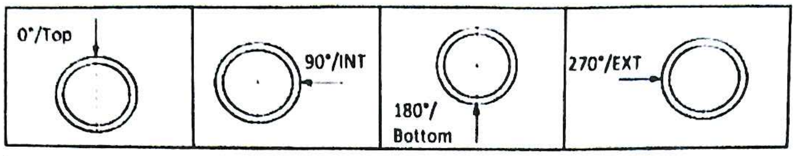
Mother Pipe Information | **Process Information**

Pipe Size (in): 8.625 x 0.322 | Grade: API-5L-X52 PSL 2 | Instrument Model No.: UCI-3000H

Serial No.: MK8821415112-1754

Probe Serial No.: MK8823417972

Key:



T= Top (Weld if Welded Material)

H= Weld Heat Affected Zone

I= Intrados of Bend

E= Extrados of Bend

TAN= Tangent of Bend

STZ= Start Transition Zone

FTZ= Finish Transition Zone

3/4= 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

Min Allowed Values: 137 | Max Allowed Values: 238

Bend ID No.	TAN T 0°	TAN H 0°	TAN 270°	STZ E 270°	STZ I 90°	MID T 0°	MID H 0°	MID E 270°	MID I 90°	FTZ E 270°	FTZ I 90°	3/4 T 0°	3/4 H 0°	3/4 EXT 270°	3/4 INT 90°
109149-1-1	194	189	199	188	182	189	180	181	186	189	185	N/A	N/A	N/A	N/A
109149-1-2	188	179	188	180	190	186	175	182	192	186	189	N/A	N/A	N/A	N/A
109149-1-3	195	188	183	179	180	203	190	182	177	183	178	N/A	N/A	N/A	N/A
109149-1-4	189	181	186	184	181	183	181	190	183	183	184	N/A	N/A	N/A	N/A
109149-2-1	192	187	190	186	184	197	190	183	183	188	187	N/A	N/A	N/A	N/A
109149-2-2	192	187	181	185	183	196	194	185	182	190	186	N/A	N/A	N/A	N/A

READINGS: ☑ MEET ACCEPTANCE CRITERIA | **PERFORMED BY:** JV | **DATE:** 6/12/2026

☐ DO NOT MEET ACCEPTANCE CRITERIA

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Attn: Firdosh Kavarana
Tulsa Tube Bending
4192 S Galveston Ave
Tulsa, OK 74107

Lab Number: 26060034
Date Received: 06/01/2026
Date Reported: 06/05/2026
PO Number: 109149-QB-1

Coupon Description: 8.625" OD x 0.322" WT
Base Material 1: API-5L-X52, ERW
Heat Number: B416745
Job Number: 109149

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -1C (Tangent)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	79	89	83	84	70	86	91	82	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -2C (Tangent Weld Seam)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse Across the Weld
Notch Location: Weld Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	12	29	52	31	27	54	70	50	35	45	85	55

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -3C (STZ-Extrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	113	104	103	107	56	63	79	66	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -4C (STZ-Intrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	114	112	119	115	75	64	69	69	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -5C (Bend Extrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	105	113	101	106	72	64	62	66	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -6C (Bend Intrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	115	120	126	120	58	66	80	68	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm
Specimen number 3 was unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.

American Piping Inspection

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Charpy V-Notch Impact Test per ASTM A370-24a

Specimen ID: -7C (Bend Weld Seam)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse Across the Weld
Notch Location: Weld Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	57	65	55	59	64	76	66	69	90	90	90	90

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

Charpy V-Notch Impact Test per ASTM A370-24a

Specimen ID: -8C (FTZ-Extrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	110	103	103	105	65	72	64	67	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -9C (FTZ-Intrados)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	135	117	125	126	69	65	71	68	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm
Specimen numbers 1 and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.

Charpy V-Notch Impact Test per ASTM A370-24a
Specimen ID: -2C (Tangent Weld Seam)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse Across the Weld
Notch Location: HAZ
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	72	93	79	81	81	80	79	80	85	100	90	92

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Charpy V-Notch Impact Test per ASTM A370-24a

Specimen ID: -7C (Bend Weld Seam)
Condition: As Received
Notch Type: V-Notch
Orientation: Transverse Across the Weld
Notch Location: HAZ
Charpy Specimen Size: 10mm x 5.0mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)				Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg
Results	108	99	139	115	64	63	69	65	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.
Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-25, Section 9.3.3.1.
Striker Radius: 8mm
Specimen number 3 was unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.

Tensile Test per ASTM A370-24a

Specimen ID: -1T (Tangent)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.318
Tensile Strength, psi	76500
Yield Strength at 0.5% EUL, psi	60900
Elongation After Fracture, %	28
YS/TS Ratio, %	0.80
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Tensile Test per ASTM A370-24a
Specimen ID: -2T (Tangent Weld Seam)
Condition: As Received
Orientation: Transverse Across the Weld
Fracture Type: Ductile

	Results
Original Width, in	0.503
Original Thickness, in	0.315
Original Area, sq in	0.1584
Maximum Load, lbf	12244
Tensile Strength, psi	77300
Location of Fracture	Base Metal

Fracture Type: Ductile

Elongation: in 2"

Tensile Test per ASTM A370-24a
Specimen ID: -3T (STZ-Extrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.299
Tensile Strength, psi	85500
Yield Strength at 0.5% EUL, psi	62400
Elongation After Fracture, %	29
YS/TS Ratio, %	0.73
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Tensile Test per ASTM A370-24a
Specimen ID: -4T (STZ-Intrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.342
Tensile Strength, psi	87900
Yield Strength at 0.5% EUL, psi	65600
Elongation After Fracture, %	28
YS/TS Ratio, %	0.75
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a
Specimen ID: -5T (Bend Extrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.306
Tensile Strength, psi	84600
Yield Strength at 0.5% EUL, psi	58900
Elongation After Fracture, %	28
YS/TS Ratio, %	0.70
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Test Report

Tensile Test per ASTM A370-24a
Specimen ID: -6T (Bend Intrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.338
Tensile Strength, psi	87600
Yield Strength at 0.5% EUL, psi	63400
Elongation After Fracture, %	30
YS/TS Ratio, %	0.72
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a
Specimen ID: -7T (Bend Weld Seam)
Condition: As Received
Orientation: Transverse Across the Weld
Fracture Type: Ductile

	Results
Original Width, in	0.503
Original Thickness, in	0.314
Original Area, sq in	0.1579
Maximum Load, lbf	13930
Tensile Strength, psi	88200
Location of Fracture	Base Metal

Fracture Type: Ductile
Elongation: in 2"

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Tensile Test per ASTM A370-24a
Specimen ID: -8T (FTZ-Extrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.300
Tensile Strength, psi	84800
Yield Strength at 0.5% EUL, psi	59300
Elongation After Fracture, %	29
YS/TS Ratio, %	0.70
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a
Specimen ID: -9T (FTZ-Intrados)
Condition: As Received
Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.345
Tensile Strength, psi	85600
Yield Strength at 0.5% EUL, psi	60100
Elongation After Fracture, %	28
YS/TS Ratio, %	0.70
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Brinell Hardness Test, ASTM E10-23
Specimen ID: -1H (Tangent)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	161

Brinell Hardness Test, ASTM E10-23
Specimen ID: -3H (STZ-Extrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	175

Brinell Hardness Test, ASTM E10-23
Specimen ID: -4H (STZ-Intrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	185

Brinell Hardness Test, ASTM E10-23
Specimen ID: -5H (Bend Extrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	177

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Brinell Hardness Test, ASTM E10-23
Specimen ID: -5H (Bend Extrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	174

Brinell Hardness Test, ASTM E10-23
Specimen ID: -6H (Bend Intrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	179

Brinell Hardness Test, ASTM E10-23
Specimen ID: -6H (Bend Intrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	179

Brinell Hardness Test, ASTM E10-23
Specimen ID: -8H (FTZ-Extrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	175

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Brinell Hardness Test, ASTM E10-23
Specimen ID: -9H (FTZ-Intrados)
Condition: As Received
Hardness Location: Surface
Hardness Test Method: ASTM E10-23 / ASTM A370-24a
Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	185

Rockwell Hardness Test
Condition: As Received
Hardness Test Method: ASTM E18-22 / ASTM A370-24a
Hardness Scale / Test Load: HRBW / 100 kgf
Specimen ID: -2H (Tangent Weld Seam)

Measurement	1
Indent Number 1	88
Indent Number 2	88
Indent Number 3	86
Indent Number 4	87
Indent Number 5	85
Indent Number 6	84
Indent Number 7	86
Indent Number 8	89
Indent Number 9	84

Test Comments:
Indent Numbers 1-3: Base Metal - 2mm from ID, Mid-Wall and 2mm from OD
Indent Numbers 4-6: HAZ - 2mm from ID, Mid-Wall and 2mm from OD
Indent Numbers 7-9: Weld Metal - 2mm from ID, Mid-Wall and 2mm from OD

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Rockwell Hardness Test

Condition: **As Received**
Hardness Test Method: **ASTM E18-22 / ASTM A370-24a**
Hardness Scale / Test Load: **HRBW / 100 kgf**
Specimen ID: **-7H (Bend Weld Seam)**

Measurement	1
Indent Number 1	88
Indent Number 2	89
Indent Number 3	90
Indent Number 4	86
Indent Number 5	88
Indent Number 6	89
Indent Number 7	86
Indent Number 8	89
Indent Number 9	86

Test Comments:
Indent Numbers 1-3: Base Metal - ID, Mid-Wall and OD
Indent Numbers 4-6: HAZ - ID, Mid-Wall and OD
Indent Numbers 7-9: Weld Metal - ID, Mid-Wall and OD

American Piping Inspection, Metallurgical Lab
18501 E. Admiral Pl.
Catoosa, OK 74015
Office: 918-266-4130

Test Report

Approved By: Jason Pierce Title: Laboratory & QA Manager

Signature:

Globe X-Ray Services, Inc.
8441 S. Union Ave.
Tulsa, OK 74132
918-446-1696

Certification # 194846

Magnetic Particle Examination Record

Customer: Tulsa Tube Bending   Date: 6/12/2026

Job/P.O. No.: 109149   Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C   Rev. No. 1/26   Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300   Serial No. 1861   Technique: X Yoke ____ Prods

Current Type: X A/C ____ D/C Magnetic Particle Type: ____ Wet X Dry ____ Fluorescent X Visible

Amperage (Prod Technique Only) ________ Amps/inch of Prod Spacing.

Part/Item Material Type: Carbon   Material Thickness: Varies

Lighting Equipment: ____ Fluorescent Light   Model # ________   Serial # ________
X White Light using Parker Research Y300 or Y400 Probe light
____ Natural White Light
____ Other: ________

Light Intensity on surface to be examined: 100 Min foot candles   N/A μW/cm²

Comments: Test The Entire Extrados (Back Half) of the Bend. From N.A to N,A, Including Weld Seam
Verify Residual Magnetism at Each End is <25 Gauss

Examination Results				
Qty:	Part Identification	Accept	Reject	Indication Description
4	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416745 Bend ID # 109149-1-1, 109149-1-2, 109149-1-3, 109149-1-4,	X		None
2	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416740 Bend ID # 109149-2-1, 109149-2-2	X		None

Examine: Jaime Vera   Qualification Level: II

Keith INDUSTRIES
20701 FM 521 Rosharon, TX 77583
+1-713-991-5670

COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TTB		ITEMS: 6 PCS
CUSTOMER PO: 26H 109149	DATE: 7/7/2026	
	PAINT SPEC: TPE_26907SP-CONSUMER ENERGY	

SURFACE PREPARATION

ITEM SALT TEST	N/A☐	RESULT: <1	SSPC SP-1 IF NEEDED		DATE	7/7/2026	7/7/2026
ABRASIVE SALT TEST	N/A☐	RESULT: <1	CLEANLINESS REQUIRED	SSPC SP-10	TIME	7:45AM	9:45AM
LABORATORY TESTING	☐YES ☒NO		CLEANLINESS ACHIEVED	☒YES☐NO	AIR TEMP	81°F	90°F
TYPE OF ABRASIVE	STEEL GRIT		PROFILE REQUIRED	2.0-4.5 MILS	RELATIVE HUMIDITY	78%	75%
NOZZLE PSI	☐ N/A 100 PSI ☒		PROFILE ACHIEVED	☒YES☐NO	STEEL	82°F	97°F
BLOTTER TEST	☒PASS		SSPC-VIS 1 USED	☒YES ☐NO	DEW POINT	73°F	81°F
DUST FREE SURFACES	☒YES ☐NO		VISUAL INSPECTION	☒YES ☐NO	STAGE OF OPERATION	PRE-BLAST	PRE-COATING

COATING APPLICATION

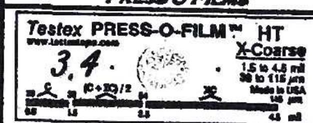
COAT 1ST	PRODUCT NAME: PIPECLAD 2000	BATCH: CH1416SW	MIN MILS: 14	TARGET PRE COAT TEMP: 450 DEG F

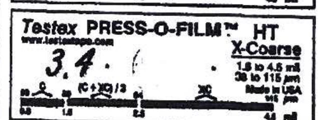
FINAL TESTING

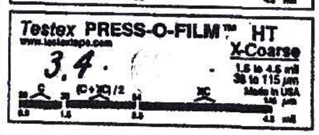
HVST @125V/MIL; 2000V MIN PASS ☒	ADHESION KNIFE TEST: PASS ☒	DFT TEST: PASS ☒	ALL REQUIREMENTS MET: PASS ☒

KI#	DESCRIPTION	MIL AVERAGE
1-1	8" X .322 30DEG BEND	27
1-2	8" X .322 30DEG BEND	26
1-3	8" X .322 30DEG BEND	28
1-4	8" X .322 30DEG BEND	25
2-1	8" X .322 45DEG BEND	27

PRESS O FILMS







KI#	DESCRIPTION	MIL AVERAGE
2-2	8" X .322 45DEG BEND	27

VERIFIED BY KEITH: ____________

Tulsa Tube Bending

Dimensional-Visual Inspection Report

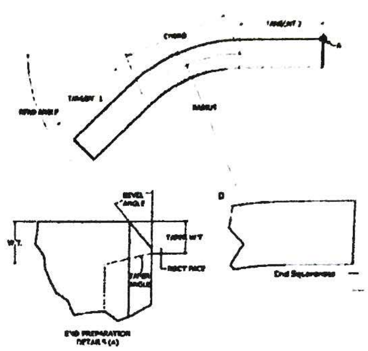
Customer:	DNOW	Pipe Size (in):	8.625 x 0.322	Report No.:	109149 DV 1 4
Customer PO No.:	662974020	Grade:	API 5L X52 PSL 2	Bend ID No.:	109149 1 4
TTB Job No.:	2611109149	Heat No.:	B416745	Code of Fabrication:	ASME B16.49 2023

Physical Dimensions

Dimension	Target	Tolerance	Actual
Tangent 1 (in)	24 MIN	MIN	26.25
Tangent 2 (in)	24 MIN	MIN	31.56
Chord (in)	20.7	n/a	20.6
Angle (deg)	30.0	+/- 0.5°	29.6
Radius, R (in)	40	+/- 1%	40.4
Performed By:	EB	Date:	6/12/2026

End Prep Dimensions

Dimension	Target	Tolerance	Actual	
Bevel Angle (deg)	37.5	+/- 2.5°	39.00	END 1
Root Face (in)	0.06	+/- 0.03 in	0.060	END 1
Min ID	7.981	+/- 0.1 in	7.940	END 1
Max ID	7.981	+/- 0.1 in	8.000	END 1
Max End Sq. (in)	0	+/- 0.09 in	0.060	END 1
Bevel Angle (deg)	37.5	+/- 2.5°	39.00	END 2
Root Face (in)	0.06	+/- 0.03 in	0.030	END 2
Min ID	7.981	+/- 0.1 in	7.900	END 2
Max ID	7.981	+/- 0.1 in	7.940	END 2
Max End Sq. (in)	0	+/- 0.09 in	0.060	END 2
Performed By:	EB	Date:	6/12/2026	



Ovality [(O.D. max – O.D. min)/O.D. nominal] x 100

Quadrant	Limit	Min OD	Max OD	% Ovality
End 1	1.0%	8.617	8.625	0.1%
Start of Bend	1.0%	8.545	8.590	0.5%
1/4 of Bend Arc	1.0%	8.495	8.580	1.0%
1/2 of Bend Arc	1.0%	8.497	8.547	0.6%
3/4 of Bend Arc	1.0%	8.530	8.599	0.8%
Stop of Bend	1.0%	8.565	8.649	1.0%
End 2	1.0%	8.604	8.640	0.4%
Performed By:	EB		Date:	6/12/2026

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.322	0.310	3.7%
1/4 of Bend Arc	10.0%	0.322	0.308	4.3%
1/2 of Bend Arc	10.0%	0.322	0.303	5.9%
3/4 of Bend Arc	10.0%	0.322	0.303	5.9%
Stop of Bend	10.0%	0.322	0.308	4.3%
Performed By:	EB		Date:	6/12/2026

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	7.742 in
Gauge Length Required =	7.981 in		

Gauge passes through the entire bend without power equipment assistance

Performed By:	MF	Date:	6/15/2026

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.
Bend is stenciled with correct information.

Performed By:	MF	Date:	6/15/2026

READINGS:

- [x] MEET ACCEPTANCE CRITERIA
- [ ] DO NOT MEET ACCEPTANCE CRITERIA